Moving Database and Log Files

Monday, October 20, 2025 12:04 PM

Moving DB files to a different drive

After taking the Database offline. Run the commands to change the SQL pointers to the new file locations, then you will need to manually move the data and log files to the new location before bringing the database online.

```
ALTER DATABASE YourDatabaseName SET OFFLINE;

ALTER DATABASE YourDatabaseName
MODIFY FILE (NAME = YourDatabaseName_Data, FILENAME = 'E:\NewLocation\YourDatabaseName_Data.mdf');

ALTER DATABASE YourDatabaseName
MODIFY FILE (NAME = YourDatabaseName_Log, FILENAME = 'E:\NewLocation\YourDatabaseName_Log.ldf');

ALTER DATABASE YourDatabaseName SET ONLINE;
```

Moving System DB files to a new drive

Permissions checklist (very important)

Ensure the **SQL Server service account** has:

- Full control on:
  - Data folders
  - Log folders
  - TempDB folders

For moving system databases you will need to stop the SQL server, perform your move(s) and pointer locations, start the SQL server back up.

Moving **model** and **msdb** data and log files follow the same procedures as above similar to moving standard database files.

For tempdb:

After stopping SQL server, run the commands below, you will need to do this for each tempdb file you have. No need to move the files as they will get recreated upon restart of the SQL server. After you have ran the commands, start SQL server again, or leave stopped until completely done if other work is needed.

```
USE master;
GO

ALTER DATABASE tempdb MODIFY FILE
(NAME = tempdev, FILENAME = 'T:\NewLocation\tempdb.mdf');
ALTER DATABASE tempdb MODIFY FILE
(NAME = templog, FILENAME = 'T:\NewLocation\templog.ldf');
GO
```

For Master:

Make sure SQL server is stopped before continuing.

Move the files in the install folder in Windows to new location…

```
C:\SQLInstallLocation\DATA\master.mdf <----- data file
C:\SQLInstallLocation\DATA\mastlog.ldf <----- log file
```

Next you'll want to change SQL startup parameters

Open SQL Server Configuration Manager → SQL Server Services → SQL Server → Properties → Startup Parameters

And modify the following parameters (DO NOT use quotes)…

```
-dD:\NewLocation\master.mdf
```

-lL:\NewLocation\mastlog.ldf

Take note of the **-l (log files) and -d (data file)**, these values NEED to be before the new file locations or the server will not start.

Optionally (but recommended) you could also move the error logs as well by altering the following parameter (DO NOT use quotes)…

-eD:\NewLocation\ERRORLOG

Again take note of the **-e (Error files**) value as that needs to be in with your path as well.

Once finished, start the SQL server back up.

Moving Databases to a new machine/server

The easiest and most efficient way to do this is to create a current backup of the database and then recover that backup on the new system.

On the source server, run the following command, replacing database name and file location where applicable…

```
BACKUP DATABASE <databasename>
TO DISK = 'D:\<backupfilelocation>\<databasename>.bak'
WITH INIT, COMPRESSION;
```

Copy the BAK file to the new server. Within SQL server run the following command, replacing any names/fields as needed…

```
RESTORE DATABASE <databasename>
FROM DISK = 'D:\<backupfilelocation>\<databasename>.bak'
WITH MOVE '<databasename>'     TO 'D:\<datafilefilelocation>\<databasename>.mdf',
   MOVE '<databasename>_log' TO 'D:\<logfilefilelocation>\<databasename>_log.ldf',
   RECOVERY;
```

Everything should recover as normal.